MODULE 3: READING, TECHNICAL DEFINITIONS & 100-WORD MASTER BANK

(CO: 4 | Weightage: 20 Marks in FAT — Question 5 Mandatory & Question 6 Choice)

Complete Study Textbook, 3-Part Architecture, Complete 100-Word Bank & IMRAD Research Deconstruction

Aligned with Official VIT BAENG101 Question Bank, Definition Blueprints, and Past FAT Papers

Scope of This Module (Official Syllabus & FAT Blueprint):
(1) Analysing Definitions: Types of Definitions (Formal, Semi-formal, Informal), (2) Methods for Expanding Definitions (Etymology, Operating Principle, Physical Description, Analogy, Exemplification), (3) Complete Master Word Bank of all 100 Official Terms, (4) Reading Comprehension & Analytical Inferences from Scientific Texts, (5) Structure and Deconstruction of a Research Article (IMRAD Framework).

1 1. VIT FAT Exam Blueprint: Questions 5 & 6 Breakdown

Module 3 governs two major 10-mark questions in your FAT exam:

Question	Exam Focus	Exact Testing Requirements & Rubric	Marks
Question 5	Technical Definitions (100-Word Bank)	Mandatory question. Five words (2M each) or ten words (1M each) are selected **strictly from the official 100-word bank**. You MUST state: Term, Class (Genus), and Differentia, followed by the complete formal sentence.	10 Marks
Question 6	Reading Comp / IMRAD Analysis	Internal choice with Question 7. Reading an unfamiliar scientific text, answering fact and inference questions, and dissecting the author's research gap and experimental claims.	10 Marks

2 2. The Architecture of Technical Definitions

2.1 2.1 The Three Types of Definitions

1. **Informal Definition:** A brief synonym or explanatory gloss placed in parentheses or apposition:
   - **Example:** "The tachometer (a rotational speed meter) recorded 3,000 RPM."
2. **Semi-Formal Definition:** An expanded phrase or relative clause clarifying meaning:
   - **Example:** "An accelerometer is a sensor which monitors dynamic acceleration forces."
3. **Formal Definition (The Exam Gold Standard):** A rigorous single-sentence statement structured according to Aristotle's classical taxonomy:

$$\text{Term} = \text{Class (Genus)} + \text{Differentia (Distinguishing Characteristics)}$$

   - **Term:** The specific technical entity being defined (e.g., **Satellite**).
   - **Class (Genus):** The broader category or family of objects to which the term belongs (e.g., **is an artificial spacecraft**).
   - **Differentia:** The precise characteristics, operating principles, or functions that distinguish this term from all other members of the same class (e.g., **placed intentionally into orbit around a planet to relay telecommunications or gather scientific telemetry data**).

2.2 2.2 The Fatal Circular Definition Trap (Zero Marks!)

A definition is circular if the differentia repeats the root word or a derivative of the term being defined. Examiners deduct all marks for circularity:

- **Fatal Circular Error:** "A **transformer** is an electrical device that **transforms** voltage." ==> **0 Marks!**
- **Correct Formal Standard:** "A **transformer** is a passive electrical device [Class] that transfers electrical energy between two or more circuits through electromagnetic induction, stepping AC voltage up or down at constant frequency [Differentia]." ==> **Full 10/10!**

2.3 2.3 Methods of Expanding Definitions

In Section B of Question 5 (or when asked to "expand one definition"), you must use one or more of these expansion methods:

1. **By Operating Principle:** Explain the physical law underlying operation (e.g. photovoltaic effect for solar cells, piezoelectric effect for quartz sensors).
2. **By Physical Description & Components:** Enumerate sub-assemblies, dimensions, and material composition.
3. **By Exemplification:** Cite industry standards, commercial models, or chemical formulas.
4. **By Comparison & Contrast:** Differentiate the item from adjacent technologies (e.g., ROM vs. RAM).

3 3. The Official 100-Word Master Definition Bank (Complete Printed Vault)

Below is the complete, unabridged catalog of all 100 terms from the official BAENG101 Word Bank. Memorize these exact three-part structures for Question 5:

S.No.	Term	Class (Genus)	Differentia & Formal Exam Definition
1	**Satellite**	An artificial celestial object or spacecraft	placed intentionally into orbit around the Earth or another planet to relay telecommunications, conduct meteorological observation, or gather scientific telemetry data. *Formal: "A satellite is an artificial spacecraft placed intentionally into orbit around a planet to relay telecommunications, conduct meteorological observation, or gather scientific telemetry data."*
2	**Equipment**	A set of physical tools, machinery, or apparatus	designed and provisioned for performing a specific task, technical operation, or laboratory experiment. *Formal: "Equipment is a set of physical tools, machinery, or apparatus designed and provisioned for performing a specific technical operation or laboratory experiment."*
3	**Wallet**	A pocket-sized flat folding case or digital software application	used to store currency, payment cards, identification documents, or cryptographic authentication keys. *Formal: "A wallet is a pocket-sized flat folding case or digital software application used to securely store currency, payment cards, or cryptographic keys."*
4	**Analysis**	A systematic method of examination	that decomposes a complex entity or dataset into its constituent elements to determine its underlying structure, behavior, and relationships. *Formal: "Analysis is a systematic method of examination that decomposes a complex entity or dataset into its constituent elements to determine its underlying structure and behavior."*
5	**Artificial Intelligence (AI)**	A subfield of computer science	dedicated to engineering software algorithms and neural networks capable of executing tasks that typically require human cognitive intelligence, such as pattern recognition, natural language reasoning, and autonomous decision-making. *Formal: "Artificial Intelligence (AI) is a subfield of computer science dedicated to engineering software algorithms and neural networks capable of executing cognitive tasks like reasoning, pattern recognition, and autonomous decision-making."*
6	**Fountainpen**	A handheld writing instrument	that utilizes capillary action and gravity to deliver water-based liquid ink from an internal refillable reservoir to a metallic nib. *Formal: "A fountainpen is a handheld writing instrument that utilizes capillary action to deliver liquid ink from an internal reservoir to a metallic nib."*
7	**Database**	A structured, electronically stored collection of data	organized systematically and managed via a Database Management System (DBMS) to enable rapid querying, indexing, updating, and transactional consistency. *Formal: "A database is a structured collection of digital information organized systematically and managed via a DBMS to enable efficient querying, indexing, and transactional integrity."*
8	**Pressure**	A scalar physical quantity	defined as the perpendicular force applied per unit surface area of an object (measured in Pascals or N/m^2). *Formal: "Pressure is a scalar physical quantity defined as the perpendicular force exerted per unit surface area of an object."*
9	**Revenue**	The total monetary gross income	generated by an enterprise through the sale of goods, provision of services, or commercial operations before deducting manufacturing and operating expenses. *Formal: "Revenue is the total gross monetary income generated by an enterprise through commercial sales or services prior to deducting operating expenses."*

10	**Drone**	An unmanned aerial vehicle (UAV)	navigated remotely by a human pilot or autonomously guided by onboard inertial sensors, GPS, and microprocessors. *Formal: "A drone is an unmanned aerial vehicle (UAV) steered remotely or guided autonomously by onboard sensors, GPS, and flight-control microprocessors."*
11	**Photosynthesis**	A biological biochemical anabolic process	by which autotrophic green plants and cyanobacteria convert light photon energy into chemical energy, synthesizing glucose from water and carbon dioxide while releasing molecular oxygen. *Formal: "Photosynthesis is a biological anabolic process in which autotrophic organisms convert light energy into chemical energy, synthesizing glucose from carbon dioxide and water with the release of oxygen."*
12	**Password**	A secret cryptographic string of characters	utilized for identity authentication to restrict unauthorized access to computer systems, networks, or encrypted resources. *Formal: "A password is a secret string of characters utilized for user authentication to verify identity and restrict unauthorized system access."*
13	**Prototype**	An early preliminary working model or sample	built to evaluate, benchmark, and iteratively validate the feasibility, ergonomics, and technical functionality of a product design prior to mass manufacturing. *Formal: "A prototype is an early working model built to benchmark, test, and validate the functionality and feasibility of a design before full-scale manufacturing."*
14	**Airline**	A commercial enterprise	that provides scheduled or chartered air transportation services for passengers and cargo utilizing certified commercial aircraft fleets. *Formal: "An airline is a commercial transportation enterprise that provides scheduled air carriage services for passengers and freight using aircraft fleets."*
15	**Recycling**	A waste management and resource recovery process	that reprocesses discarded municipal or industrial materials into new usable products or raw feedstocks to diminish virgin resource extraction and pollution. *Formal: "Recycling is a waste recovery process that converts discarded solid waste materials into reusable raw commodities to mitigate resource depletion and environmental degradation."*
16	**Microphone**	An acoustic electroacoustic transducer	that converts sound pressure wave oscillations in air into corresponding analog or digitized electrical audio signals. *Formal: "A microphone is an electroacoustic transducer that converts sound pressure wave oscillations into proportional electrical audio signals."*
17	**Optimist**	An individual characterized by a psychological disposition	that anticipates favorable results, positive outcomes, and constructive growth even in adverse or uncertain circumstances. *Formal: "An optimist is an individual characterized by a cognitive disposition that consistently expects favorable outcomes and constructive solutions despite adversity."*
18	**UPI (Unified Payments Interface)**	An instantaneous real-time payment architecture	developed by NPCI in India that facilitates immediate inter-bank peer-to-peer and peer-to-merchant fund transfers via mobile devices without exposing bank account numbers. *Formal: "UPI is a real-time mobile payment architecture that facilitates instant inter-bank monetary transactions without revealing sensitive account credentials."*
19	**Cybersecurity**	The technical and operational discipline	focused on defending networked computer hardware, software applications, communication protocols, and sensitive digital data from unauthorized infiltration, cyberattacks, and exploitation. *Formal: "Cybersecurity is the discipline focused on protecting computing systems, digital networks, and electronic data from unauthorized intrusion, damage, or cyberattacks."*
20	**Portfolio**	A curated systematic collection	of financial investments, professional projects, design artifacts, or academic works demonstrating an individual's or institution's competencies and assets. *Formal: "A portfolio is a curated collection of financial assets, professional accomplishments, or creative artifacts demonstrating competence and value."*
21	**Cement**	A fine mineral binder and hydraulic construction material	manufactured by calcining limestone and clay which sets, hardens, and adheres to aggregate stones upon hydration

			with water to form concrete. *Formal: "Cement is a fine hydraulic mineral binder made from calcined limestone that chemically hydrates with water to bind aggregates into solid concrete."*
22	**Browser**	A client-side software application	used to retrieve, render, and navigate hypertext markup, multimedia assets, and web applications hosted across the World Wide Web via HTTP/HTTPS protocols. *Formal: "A browser is a software application that interprets HTML/CSS and fetches web resources over HTTP/HTTPS to render interactive web pages for users."*
23	**Solar panel**	An integrated photovoltaic assembly of solar cells	that directly converts incident solar electromagnetic radiation into direct-current (DC) electricity via the internal photovoltaic effect. *Formal: "A solar panel is an assembly of interconnected photovoltaic cells that converts solar radiant energy directly into direct-current electrical energy."*
24	**Biography**	A non-fiction written narrative account	of an individual's life history, career milestones, achievements, and legacy, authored by an external biographer. *Formal: "A biography is a non-fictional literary work detailing the life history, achievements, and legacy of an individual, composed by another person."*
25	**Battery**	An electrochemical energy storage device	consisting of one or more galvanic cells that convert stored chemical energy into electrical energy through spontaneous redox reactions. *Formal: "A battery is an electrochemical storage device comprising galvanic cells that converts stored chemical energy into electrical current via redox reactions."*
26	**Quick commerce**	A high-speed e-commerce logistics and fulfillment model	designed to deliver low-weight FMCG groceries and essential commodities to consumers within 10 to 30 minutes utilizing hyper-local dark stores. *Formal: "Quick commerce is an e-commerce fulfillment model that delivers essential consumer goods within 10 to 30 minutes using distributed hyper-local micro-warehouses."*
27	**Sensor**	An input electrical or electronic transducer	that detects a physical parameter (such as temperature, light, pressure, or motion) and transforms it into a readable, quantifiable electrical signal. *Formal: "A sensor is an electronic transducer that detects environmental physical stimuli and translates them into measurable electrical signals."*
28	**Restaurant**	A commercial hospitality establishment	where prepared culinary food and beverages are sold and served to patrons for on-premises consumption or takeaway delivery. *Formal: "A restaurant is a commercial establishment where culinary food and beverages are prepared, purchased, and served to dining customers."*
29	**3D Printing**	An additive manufacturing process	that synthesizes three-dimensional physical objects layer upon layer directly from a 3D digital CAD model using polymers, resins, or metallic powders. *Formal: "3D printing is an additive manufacturing fabrication technique that deposits successive layers of material guided by a 3D computer CAD file to build physical objects."*
30	**Folder**	A graphical user interface container or physical cardboard binder	employed to categorize, group, and store digital files or paper documents systematically for streamlined retrieval. *Formal: "A folder is a physical binder or digital directory container employed to organize and store related documents systematically."*
31	**Pollution**	The introduction of contaminants or harmful substances	into the natural biosphere (air, water, or soil) causing ecological disruption, physiological harm to organisms, or environmental degradation. *Formal: "Pollution is the contamination of the physical environment by xenobiotic or harmful substances that cause adverse ecological and biological consequences."*
32	**Aircraft**	A heavier-than-air or lighter-than-air vehicle	capable of sustained flight through atmospheric air via aerodynamic lift generated from fixed wings, rotating airfoils, or buoyant aerostatic gas. *Formal: "An aircraft is any vehicular machine capable of sustained atmospheric flight supported by aerodynamic lift or aerostatic buoyancy."*
33	**Emoji**	A standardized digital ideogram or pictogram	embedded in electronic communication to express emotional states, gestures, concepts, or objects in informal text-based messaging. *Formal: "An emoji is a standardized visual*

			digital pictogram used in electronic text communication to convey emotion, tone, or physical objects."
34	**Software**	A collection of computer programs, scripts, and executable instructions	that direct a computer processor to perform specific data processing, hardware management, or computational tasks. *Formal: "Software is a collection of programmed machine-readable instructions and documentation that directs computing hardware to execute defined operations."*
35	**Tax**	A compulsory financial levy or fiscal charge	imposed by a governmental authority on individuals, corporate transactions, or assets to fund public infrastructure and state expenditures. *Formal: "A tax is a compulsory fiscal levy imposed by a sovereign government on citizens and commercial entities to fund public services and infrastructure."*
36	**Smartphone**	A handheld cellular mobile communications device	integrating a high-resolution touchscreen, mobile operating system, internet connectivity, and hardware sensors to run diverse software applications. *Formal: "A smartphone is a handheld mobile computing device that combines cellular telephony with computing capabilities, internet access, and multi-sensor touchscreen hardware."*
37	**Convocation**	A formal university academic assembly and ceremony	convened to officially confer collegiate degrees, diplomas, and honors upon graduating students in the presence of academic dignitaries. *Formal: "A convocation is a formal academic ceremony held by a university to officially confer earned degrees and honors upon its graduates."*
38	**Airdrop**	A wireless proprietary file-sharing protocol or supply drop	that transmits digital files peer-to-peer over close-range Wi-Fi and Bluetooth without an active external internet connection. *Formal: "Airdrop is a close-range wireless transfer protocol that enables encrypted peer-to-peer transmission of digital files between adjacent devices."*
39	**Microscope**	An optical or electronic laboratory instrument	employing lenses or electron beams to produce highly magnified, high-resolution visual images of microscopic specimens imperceptible to the naked eye. *Formal: "A microscope is an optical or electron instrument designed to produce magnified images of minute biological specimens or crystalline structures."*
40	**Phishing**	A deceptive social engineering cyberattack	wherein an adversary masquerades as a trustworthy entity via email or spoofed websites to induce victims into disclosing passwords, credit card credentials, or 2FA tokens. *Formal: "Phishing is a fraudulent cyberattack where attackers impersonate legitimate organizations to trick victims into revealing sensitive authentication credentials."*
41	**Bionote**	A concise professional biographical profile	written in the third person (typically 50-100 words) highlighting an author's, researcher's, or speaker's credentials, current role, and prominent achievements. *Formal: "A bionote is a concise, third-person biographical summary highlighting a professional's credentials, institutional affiliation, and notable achievements."*
42	**Communication**	A dynamic reciprocal transmission process	where information, ideas, messages, or emotions are exchanged between a sender and receiver through verbal, non-verbal, or electronic mediums. *Formal: "Communication is the reciprocal process of transferring information and shared meaning between a sender and a recipient through symbolic codes or mediums."*
43	**Cylinder**	A fundamental three-dimensional geometric solid	bounded by two congruent parallel circular bases connected by a continuous curved lateral surface equidistant from a central axis. *Formal: "A cylinder is a geometric three-dimensional solid formed by two congruent, parallel circular bases joined by a curved surface."*
44	**Virtual Reality (VR)**	A computer-generated immersive simulation technology	that replicates a realistic three-dimensional interactive environment experienced through head-mounted displays and sensory tracking hardware. *Formal: "Virtual Reality (VR) is an interactive computer-generated technology that immerses the user within an artificial 3D simulated environment via specialized head-mounted displays."*

45	**Volunteer**	An individual	who freely renders time, labor, or services for charitable, humanitarian, or community causes without financial compensation or legal coercion. *Formal: "A volunteer is a person who willingly undertakes humanitarian or community services without remuneration or contractual obligation."*
46	**Patent**	A sovereign legal intellectual property grant	conferring exclusive commercial monopoly rights to an inventor over a novel, non-obvious, and industrially applicable invention for a designated period (usually 20 years). *Formal: "A patent is an exclusive legal intellectual property right granted by a government to an inventor for a novel, useful invention for a term of 20 years."*
47	**Internet banking**	An electronic financial banking platform	allowing customers to perform retail banking operations, bill payments, and fund transfers remotely via a secure web portal without visiting a physical branch. *Formal: "Internet banking is a web-based banking facility enabling account holders to execute financial transactions and fund transfers remotely via secure portals."*
48	**Dessert**	A sweet culinary confectionery course	traditionally served at the conclusion of a formal dinner or lunch meal (e.g., pastries, fruits, custards, or confections). *Formal: "Dessert is a sweet culinary dish served as the culminating course of a formal meal."*
49	**Authentication**	A security verification protocol	that confirms the claimed identity of a user, workstation, or data packet through verification of factors like passwords, biometrics, or hardware tokens. *Formal: "Authentication is the security process of verifying that a system user or entity is genuinely who they purport to be through credential validation."*
50	**Podcast**	An episodic digital audio program	streamed or downloaded over the internet via RSS syndication feeds for playback on personal media devices. *Formal: "A podcast is an episodic series of spoken-word digital audio recordings distributed online for asynchronous user streaming or downloading."*
51	**Renewable Energy**	Usable power derived from natural replenishment sources	that are naturally replenished on a human timescale (such as solar radiation, wind kinetics, hydroelectric flow, geothermal heat, and biomass) without depleting finite reserves. *Formal: "Renewable energy is power harnessed from naturally recurring environmental phenomena (solar, wind, hydro) that are replenished indefinitely without exhaustible depletion."*
52	**Clog**	An obstruction or heavy wooden footwear	that impedes the continuous fluid flow through a pipe, conduit, or drainage canal due to accumulated solid debris. *Formal: "A clog is an accumulated physical blockage that impedes or halts the fluid flow through an engineering pipe or channel."*
53	**Data**	Unprocessed raw facts, statistics, or qualitative observations	collected for reference, statistical analysis, or algorithmic computational processing into structured actionable information. *Formal: "Data consists of raw, uninterpreted factual symbols, values, or measurements collected for computational processing and analysis."*
54	**Resumé**	A formal professional career document	summarizing an applicant's education, employment history, technical skill set, and accomplishments to secure an interview. *Formal: "A resumé is a structured, concise professional document summarizing a job applicant's educational background, work experience, and technical competencies."*
55	**Projector**	An optical display output peripheral device	that projects magnified video, computer imagery, or presentation slides onto a large reflective screen or flat surface using high-lumen light sources. *Formal: "A projector is an optoelectronic display device that projects computer-generated visual media onto a large projection surface."*
56	**Biometric Authentication**	An identity verification cybersecurity mechanism	that validates user identity through automated analysis of unique physiological traits (fingerprints, facial geometry, retina/iris scans) or behavioral markers. *Formal: "Biometric authentication is an identity verification technique that identifies individuals based on distinct physiological characteristics such as fingerprints or retinal patterns."*

57	**Pessimist**	An individual possessing a cognitive disposition	that habitually anticipates undesirable outcomes, failures, or the most unfavorable aspects of any given situation. *Formal: "A pessimist is an individual with a psychological tendency to anticipate failure, disadvantageous consequences, and adverse developments."*
58	**Walkalator**	A motorized passenger conveyance system	consisting of a flat, continuous, horizontal or slightly inclined moving beltway designed to transport pedestrians across lengthy transit terminals and concourses. *Formal: "A walkalator is a continuous power-driven moving horizontal walkway designed to convey pedestrians across long airport or transit corridors."*
59	**Circuit**	A closed conducting loop or interconnection	consisting of active and passive electrical components providing an uninterrupted path for continuous electric current flow. *Formal: "A circuit is an interconnected closed loop of electrical components through which electric charge can continuously flow under a potential difference."*
60	**Celebrity**	A high-profile person	who commands widespread public recognition, fame, and media scrutiny within society due to achievements in entertainment, sports, or public life. *Formal: "A celebrity is a prominent public figure who commands extensive social recognition and media attention due to public accomplishments."*
61	**USB drive**	A portable solid-state flash memory data storage peripheral	integrating a Universal Serial Bus connector to enable rewritable plug-and-play data interchange across computing platforms. *Formal: "A USB drive is a portable solid-state flash memory device fitted with a USB interface for rewritable digital data storage and transfer."*
62	**Innovation**	The practical implementation and deployment	of a novel idea, methodology, or technology that creates demonstrable value, efficiencies, or functional improvements in existing processes. *Formal: "Innovation is the practical introduction of novel methods, inventions, or workflows that significantly elevate technical efficiency, value, or utility."*
63	**Telemedicine**	A remote medical clinical healthcare discipline	that utilizes telecommunications and digital video platforms to provide remote patient diagnosis, health monitoring, and medical consultations across physical distances. *Formal: "Telemedicine is the delivery of medical healthcare and clinical diagnosis across distances using real-time digital telecommunication technologies."*
64	**Username**	A unique alphanumeric identifier string	assigned to or selected by an individual to identify their specific user account on a computing system or digital service. *Formal: "A username is a unique alphanumeric string used to identify and distinguish a specific user account within an electronic network."*
65	**Solar eclipse**	An astronomical celestial alignment event	occurring when the Moon passes directly between the Sun and Earth, thereby fully or partially obstructing the Sun's light from reaching the terrestrial surface. *Formal: "A solar eclipse is an astronomical event occurring when the Moon traverses between the Sun and Earth, temporarily casting its shadow on the Earth."*
66	**Malware**	Malicious executable software code	intentionally engineered to compromise, damage, disrupt, or gain illicit unauthorized access to computer systems and confidential digital networks. *Formal: "Malware is malicious software engineered to infiltrate, damage, or exploit vulnerable computer systems without the user's consent."*
67	**Square**	A two-dimensional regular quadrilateral polygon	possessing four sides of equal length and four internal right angles measuring exactly 90 degrees. *Formal: "A square is a planar regular quadrilateral with four congruent sides and four internal right angles."*
68	**Headphones**	A personal electroacoustic listening device	featuring a pair of miniature audio speakers worn over or inside the user's ears to reproduce audio privately. *Formal: "Headphones are a pair of small electroacoustic audio drivers worn over or inside the ears for private sound listening."*
69	**Autobiography**	A non-fiction self-authored literary work	in which a person writes a chronological and introspective account of their own life experiences, career, and reflec-

			tions. *Formal: "An autobiography is a self-authored literary narrative chronicling the writer's own life experiences and personal reflections."*
70	**Smart watch**	A wearable wrist-mounted computing device	featuring a touchscreen display, biometric sensors (heart rate, SpO2, accelerometer), and wireless smartphone connectivity. *Formal: "A smart watch is a wearable wrist device that combines digital timekeeping with biometric health tracking and wireless smartphone synchronization."*
71	**Debugging**	A software engineering diagnostic troubleshooting procedure	that systematically isolates, traces, and eliminates software coding bugs, syntax errors, and logical discrepancies from source code. *Formal: "Debugging is the systematic engineering process of detecting, isolating, and rectifying defects and logic errors within computer software."*
72	**Hyperlink**	An interactive digital electronic reference link	embedded in a hypertext document that redirects the user directly to another file, webpage, or specific coordinate when clicked. *Formal: "A hyperlink is a clickable reference anchor in a digital document that automatically navigates the user to a targeted external or internal resource."*
73	**Dias**	A low elevated wooden or stone platform	installed at the front of an auditorium or hall to elevate dignitaries, speakers, or performers above the general audience. *Formal: "A dias is an elevated platform positioned in an auditorium or assembly hall to honor or provide visibility for speakers and dignitaries."*
74	**Emoticon**	A typographic facial representation	constructed from typographical punctuation symbols and characters (e.g., :-) or :-D) to signify emotional nuance in text communication. *Formal: "An emoticon is a typographic sequence of punctuation glyphs used to represent human facial expressions and tone in written messaging."*
75	**Sustainability**	An ecological and socio-economic governance paradigm	that meets present societal and economic requirements without compromising the capacity of future generations to meet their own resource needs. *Formal: "Sustainability is the practice of consuming resources responsibly to meet present human needs without impairing the ecological capacity of future generations."*
76	**Calculator**	An electronic or mechanical calculating appliance	designed to execute arithmetic operations, scientific functions, and algebraic computations with prompt algorithmic accuracy. *Formal: "A calculator is a dedicated electronic device engineered to perform mathematical and scientific calculations rapidly."*
77	**Tournament**	A structured athletic or academic sports competition	involving multiple contested rounds and elimination stages to determine a definitive supreme winner or champion. *Formal: "A tournament is an organized series of competitive matches in which multiple participants or teams contest successive stages to determine a champion."*
78	**Philanthropist**	An affluent benevolent individual	who systematically donates financial wealth, time, and resources to promote human welfare, educational institutions, and charitable causes. *Formal: "A philanthropist is a charitable individual who donates substantial personal capital and resources to elevate human welfare and social progress."*
79	**3D Printing (Additive)**	A precision digital rapid-prototyping manufacturing technology	that constructs three-dimensional solid geometries through iterative computer-controlled deposition of fused materials layer by layer. *Formal: "3D printing is a computerized fabrication method that creates solid physical objects by building consecutive layers of material based on digital models."*
80	**Revenue (Commercial)**	The total aggregate financial turnover	derived from operational product sales, royalty accruals, and client invoices prior to any deductions for cost of goods sold. *Formal: "Revenue is the aggregate gross financial proceeds earned by an organization through its primary business operations and service deliveries."*
81	**Airdrop (Military/Logistical)**	An aerial logistical delivery maneuver	where military supplies, humanitarian aid, or personnel are discharged from flying aircraft using aerodynamic parachutes into remote zones. *Formal: "An airdrop is an aviation delivery operation that drops emergency humanitarian relief or military provisions to the ground via parachute."*

82	**Material**	A physical substance or composite matter	possessing specific mechanical, thermal, electrical, and structural properties utilized to fabricate industrial components and artifacts. *Formal: "A material is any tangible substance with defined physical and chemical properties utilized for engineering fabrication and synthesis."*
83	**Process**	A systematic series of sequential operations or steps	executed in a prescribed order to transform input materials or raw parameters into a designated end-product or outcome. *Formal: "A process is a structured sequence of interconnected activities or chemical steps designed to transform raw inputs into a finished output."*
84	**Method**	An established systematic procedure or technique	employed to accomplish a designated analytical, scientific, or practical objective with repeatable precision. *Formal: "A method is a standardized, repeatable procedure executed to solve a specific scientific problem or accomplish a practical technical task."*
85	**System**	An interconnected network of interacting entities or components	functioning collaboratively according to established rules to accomplish a cohesive, overarching unified objective. *Formal: "A system is an organized set of interacting physical or logical components operating collectively to fulfill a designated common objective."*
86	**Device**	A physical piece of equipment or mechanism	engineered and fabricated to perform a specific dedicated functional purpose, often powered electrically or mechanically. *Formal: "A device is an engineered instrument, appliance, or mechanism designed to execute a defined physical or computational function."*
87	**Structure**	The geometric physical arrangement and organization	of interconnected structural elements or molecules that withstands applied loads and establishes spatial stability. *Formal: "A structure is the organized spatial arrangement of interconnected supporting elements designed to bear loads and maintain stable equilibrium."*
88	**Function**	The assigned operational purpose or mathematical mapping	that defines the specific utility, role, or input-output relationship executed by a system component or algorithm. *Formal: "A function is the dedicated operational role, duty, or algorithmic mapping executed by an engineering mechanism or mathematical entity."*
89	**Component**	A modular constituent part or sub-assembly	that integrates alongside other modular units to assemble a larger functioning machine, electronic system, or software application. *Formal: "A component is an individual modular constituent part that integrates with other elements to form a complete functioning apparatus."*
90	**Measurement**	A quantitative physical experimental evaluation	that determines the magnitude, dimension, capacity, or quantity of an unknown variable compared against a standardized international SI reference unit. *Formal: "Measurement is the process of quantifying a physical dimension or property by comparing it against an established standard unit of evaluation."*
91	**Design**	A creative engineering specification and planning process	that formulates blueprints, architectural schematics, and functional parameters to solve a practical technical challenge. *Formal: "Design is the systematic engineering process of conceptualizing, formulating, and specifying the structure and parameters of a solution to fulfill defined requirements."*
92	**Testing**	An empirical verification and validation procedure	carried out to benchmark the reliability, performance, safety margins, and specification compliance of a component or software program under stressed conditions. *Formal: "Testing is an empirical experimental procedure executed to verify whether a system meets specified performance criteria and functions reliably."*
93	**Research**	A disciplined, methodical intellectual investigation	conducted to discover new empirical facts, formulate scientific hypotheses, expand academic knowledge, and resolve technical challenges. *Formal: "Research is a disciplined systematic inquiry aimed at discovering new empirical principles, revising accepted theories, or solving technological problems."*
94	**Experiment**	A controlled empirical scientific test or trial	conducted by intentionally manipulating independent variables while monitoring dependent variables to test an empirical hypothesis. *Formal: "An experiment is a controlled*

			empirical scientific procedure designed to test a hypothesis by manipulating variables under monitored conditions."
95	**Energy**	A fundamental scalar physical capacity	that quantifies the ability of a system to perform work, exert force, or liberate thermal heat (governed by the Law of Conservation of Energy, measured in Joules). *Formal: "Energy is a fundamental scalar physical property that quantifies the quantitative capacity of a system to perform work or produce heat."*
96	**Temperature**	A macroscopic thermodynamic state variable	that measures the average kinetic energy of molecular microscopic motion within a substance (measured in Kelvin or Celsius). *Formal: "Temperature is a thermodynamic state variable that measures the average microscopic translational kinetic energy of constituent particles in matter."*
97	**Voltage**	The electric potential difference	between two spatial points in an electric field representing the work needed per unit electric charge to displace a test charge between them (measured in Volts). *Formal: "Voltage is the difference in electrical potential between two points, denoting the work required per unit charge to transport electric current."*
98	**Network**	An interconnected infrastructure of communication nodes	linked through physical transmission cables or wireless channels to exchange data packets and share computational resources via common protocols. *Formal: "A network is an interconnected group of autonomous computer nodes linked by communication media to exchange digital data and share computing resources."*
99	**Storage**	A persistent or volatile electronic computer memory medium	engineered to record, retain, and retrieve digital bytes and program binaries indefinitely or during active power delivery. *Formal: "Storage is a computer hardware medium engineered to retain digital binary information and data files for subsequent computation."*
100	**Installation**	The physical commissioning or software configuration procedure	wherein an apparatus, machinery, or software package is mounted, connected, and initialized into operational readiness within an environment. *Formal: "Installation is the technical process of setting up, mounting, and configuring an apparatus or software system into active operational readiness."*

4 4. Reading Comprehension & Structure of a Research Article (IMRAD)

4.1 4.1 The IMRAD Framework (IEEE Standard)

Every scientific and engineering research paper adheres strictly to the four-part IMRAD architecture:

- **I — Introduction:** What problem was investigated, what is the existing scholarship, and what is the specific **Research Gap** and thesis?
- **M — Methods:** How was the study conducted? Detailed apparatus specifications, sample preparation, numerical parameters, and reproducible protocols.
- **R — Results:** What empirical facts were discovered? Unbiased statistical data, stress-strain curves, XRD patterns, without subjective speculation.
- **D — Discussion:** What do the results mean? Interpretation of mechanisms, comparison with prior published findings, limitations, and future research recommendations.

4.2 4.2 Drawing Analytical Inferences vs. Explicit Facts

- **Explicit Fact:** Directly stated in the text (e.g. "The reaction occurred at 85 °C").
- **Analytical Inference:** A logical conclusion deduced from synthesis of data points (e.g. "Because yield peaked at 85 °C and plummeted at 95 °C, the enzyme experiences thermal denaturation above 85 °C").

5 5. Worked FAT Question Bank Problems (Full 10/10 Model Solutions)

5.1 Problem 3.1: Full FAT Question 5 Model Answer (10 Marks)

Question: Write formal three-part definitions for any FIVE of the following terms selected from the official word bank, clearly identifying the Term, Class, and Differentia: (a) Drone, (b) Cybersecurity, (c) 3D Printing, (d) Phishing, (e) Renewable Energy.

Model 10/10 Answer:

1. Drone:
- **Term:** Drone
- **Class:** An unmanned aerial vehicle (UAV)
- **Differentia:** steered remotely by a human pilot or guided autonomously by onboard inertial sensors, GPS, and flight-control microprocessors.
- **Formal Definition:** A drone is an unmanned aerial vehicle (UAV) steered remotely or guided autonomously by onboard sensors, GPS, and microprocessors.

2. Cybersecurity:
- **Term:** Cybersecurity
- **Class:** The technical and operational discipline
- **Differentia:** focused on defending networked computer hardware, software applications, and sensitive digital data from unauthorized infiltration, cyberattacks, and exploitation.
- **Formal Definition:** Cybersecurity is the technical discipline focused on defending computing systems, networks, and electronic data from unauthorized intrusion or cyberattacks.

3. 3D Printing:
- **Term:** 3D Printing
- **Class:** An additive manufacturing fabrication process
- **Differentia:** that constructs three-dimensional physical objects layer upon layer directly from a 3D digital CAD model using polymers, resins, or metallic powders.
- **Formal Definition:** 3D printing is an additive manufacturing fabrication technique that deposits successive layers of material guided by a 3D CAD model to build solid physical objects.

4. Phishing:
- **Term:** Phishing
- **Class:** A deceptive social engineering cyberattack
- **Differentia:** wherein an adversary masquerades as a trustworthy entity to trick victims into disclosing confidential passwords, credit card credentials, or authentication tokens.
- **Formal Definition:** Phishing is a fraudulent cyberattack where attackers impersonate legitimate entities to deceive victims into disclosing sensitive authentication credentials.

5. Renewable Energy:
- **Term:** Renewable Energy
- **Class:** Usable power derived from natural replenishment sources
- **Differentia:** that are naturally replenished on a human timescale (such as solar, wind, and hydro) without exhaustible reserve depletion.
- **Formal Definition:** Renewable energy is power harnessed from recurring environmental phenomena that are replenished indefinitely without resource depletion.